

Software Quality Factors Analysis

I initially wanted to test a C++ desktop app, created with Qt, but I encountered an immense number of errors when trying to use metric tools on that project. I tested at least 5 tools and none of them worked, so I decided to keep the same software quality factors but change the project. I will be testing a Web app with a Java Spring BE and a React FE.

The Quality Model I've chosen is the **ISO 9126** from 2001

Functionality	Reliability	Usability	Efficiency	Maintainab.	Portability
Suitability	Maturity	Understand	Time behav.	Analysability	Adaptability
Accurateness	Fault tolerance	Learnability	Resource beh.	Changeability	Instability
Interoperability	Recoverability	Operability		Stability	Conformance
Compliance				Testability	Replaceability
Security					

The factors I've chosen are:

- **Usability** - the degree to which a software is easy to use.
- **Maintainability** - the ease with which the software can be modified.
- **Portability** - the ease with which a software can be migrated from one environment to the other.
- **Functionality** - degree to which software satisfies stated needs.

This is how I to measured them:

Usability - The plan was:

- I will define some usability criteria such as "X out of Y users must be able to perform task Z in a certain amount of time or with a certain number of errors (errors resulted from a lack of system understanding and not software errors)(like clicking the share icon instead of copy because you don't understand what the symbols mean)"
- Then I will have a few users that match the target user type for the application, test it.
- I will observe how they react while they use the app (if they feel frustrated or not).
- And then I will interview them about the app, making sure they know I am evaluating the app and NOT their performance, so that they feel comfortable talking freely about the app's performance.

For a limited number of functionalities (limited so it's easier to test) I have defined the following criteria which I expect at least a specified amount of users to meet in order for the app to pass the Usability test.

Functionality	Criteria
Login & Find internships by work type and salary	3 out of 3 users should login, find the page and filter it under 30 seconds with no errors
Apply for an internship and view your current applications	3 out of 3 users should apply within 20 seconds from the internships page and find the page to view their application within another 30 seconds. All with maximum one error
Identify key informations about an internship (eg: salary, location, required knowledge)	3 out of 3 users should identify such information with no error on the internship page
Sign up as a company and create a new internship offer	2 out of 3 users should add a company with no errors and 2 out of 3 users should be able to create an internship, all within 1 minute & 30secs
Find how many ongoing internships a company has	3 out of 3 should find this information within 15 seconds of adding a new internship

Usability - Results

Functionality	User	Time to complete (min.seconds)	Notes
Find internships by work type and salary	User 01	1:00	There were complaints on the style The filter buttons is not big enough to be noticed easily
	User 02	0:29	Enter doesn't work, must use the button. Can't see applied filters
	User 03	1:20	-

Apply for an internship and view your current applications	User 01	1:38	After application there should be a pop-up saying if it was successful or not.
	User 02	0:49	Couldn't upload a long resume, no error message.
	User 03	1:10	Can't change the uploaded CV
Identify key informations about an internship (eg: salary, location, required knowledge)	User 01	0:31	-
	User 02	0:31	intuitive
	User 03	0:20	-
Sign up as a company and create a new internship offer	User 01	1:20	Logout doesn't have a confirmation pop-up. bug: browser alert instead of pop-up when adding internship. Company register is very straightforward.
	User 02	1:09	The alert is not great
	User 03	2:10	Would be nice to auto redirect to the new internship
Find how many ongoing internships a company has	User 01	0:9	bug: no message shows up where there are no users for the current internship. The dashboard could be a sidebar (it would be more intuitive).
	User 02	0:13	The buttons do nothing
	User 03	0:10	-

Interpretation

- The app is intuitive, users did not get stuck anywhere and did not encounter any major errors when using the software. The interface was self explanatory enough that the users did not need to ask for any help or guidance in executing the given steps.

- Below are the results correlated with the initial criteria. Out of 5 criterias, 4 were met, therefore the app has a high usability rating.

Functionality	Criteria	Results
Login & Find internships by work type and salary	3 out of 3 users should login, find the page and filter it under 30 seconds with no errors	30 seconds was way to fast, but most of the users did it in around a 1:30 with no errors
Apply for an internship and view your current applications	3 out of 3 users should apply within 20 seconds from the internships page and find the page to view their application within another 30 seconds. All with maximum one error	Yes, users achieved this!
Identify key informations about an internship (eg: salary, location, required knowledge)	3 out of 3 users should identify such information with no error on the internship page	Yes, they said the information is very intuitive and easy to find!
Sign up as a company and create a new internship offer	2 out of 3 users should add a company with no errors and 2 out of 3 users should be able to create an internship, all within 1 minute & 30 secs	2 out of 3 users did it in under 90 seconds!
Find how many ongoing internships a company has	3 out of 3 should find this information within 15 seconds of adding a new internship	The average was 10 seconds!

- Based on this test, the currently implemented functionalities of the app are user friendly.

Portability - The plan was:

- Test if all the app's functionalities work on different types of browser (chromium based vs not chromium based), and how it works on different devices (mobile vs desktop)

Portability - Results

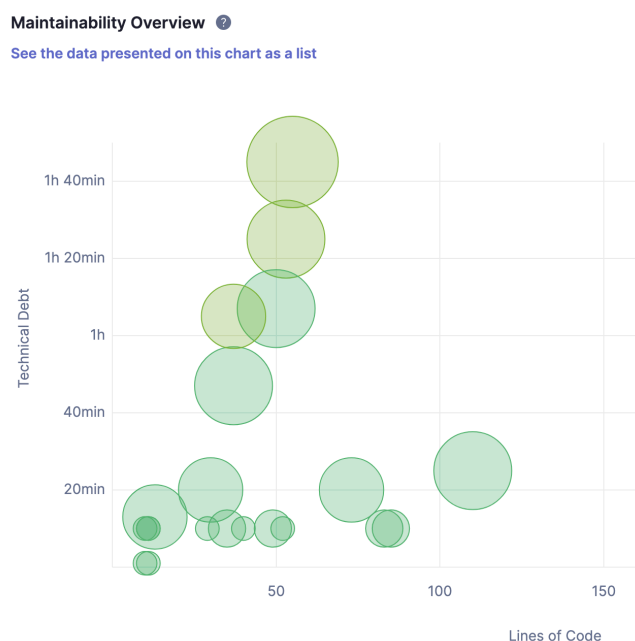
- There were minimal changes in looks between Firefox and Chromes (mostly regarding the shape of the scroll bars and the default look of some text fields)
- The app is not mobile friendly, the interface gets distorted on smaller screens and is unreadable.
- Therefore, considering this is a browser app it is highly portable on desktop devices, it will work on all browsers (with minor visual differences). But because it is a browser app users might expect it to be openable on mobile, which is not. So the app does decently in the portability department but it is not perfect.

Maintainability - The plan was:

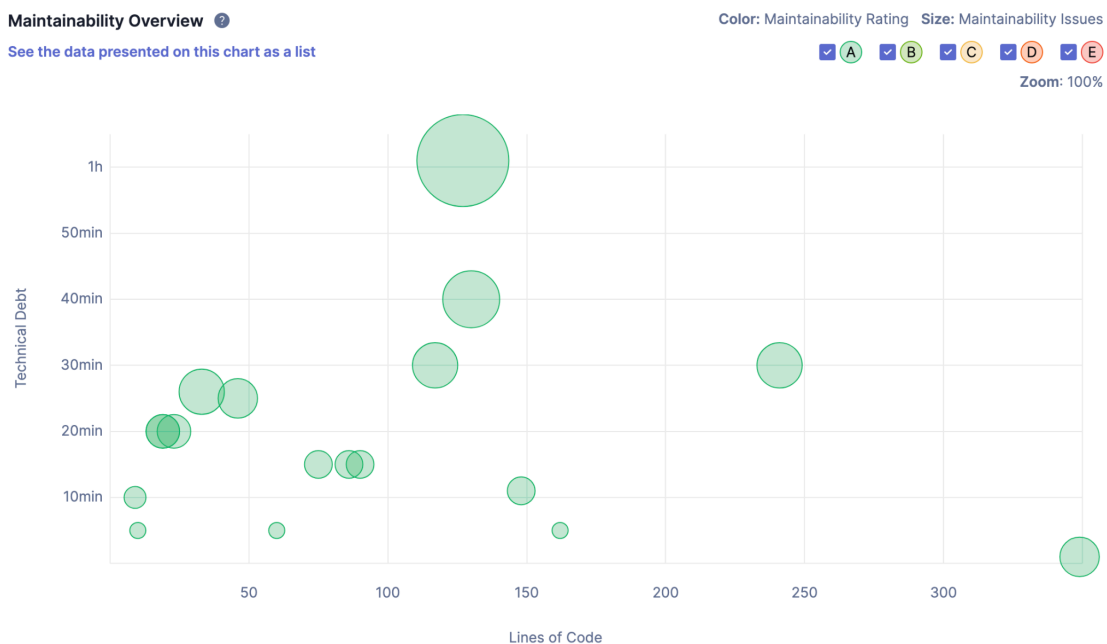
- Use Sonar to measure maintainability metrics.

Maintainability - Results

- For the backend code the diagram is below. Most of these problems come from code not covered by tests, the use of some wildcards, comments forgotten in the code, unused imports or deprecated use of annotations. These are all understandable but still, not a real cause for concern for right now.
- Considering this app is well divided into modules and classes, I believe this backend is highly maintainable.



- For the frontend code the diagram is below. The issues found are: unused imports, forgotten comments or todos, lack of props validation. These are all easy fixes. Sonar estimates around 5 hours and I would have to agree. None of the issues found are cause for any real problems.
- As for the architecture, the frontend is also very split into separate functions for each thing. With such a good separation of concerns, this code is definitely highly maintainable.



Functionality - The plan was:

- Measuring how many of the original requirements have been met
- Check how many defects (bugs) the app has per functionality?

Functionality - Results

- Below are all the original requirements. 19 out of 45 initial requirements were implemented. This is approximately 42%. Not even half. Many of the not implemented requirements were crucial to the app, and the app can NOT be used without those. Therefore the functionality score for this app is on the low side.

App functionalities

The system shall allow users to register.	
The system shall display a list with all available offers for internship programs.	
The system shall display all the necessary steps to apply for each individual internship program.	
The system shall display a short description about the offered internship program for each individual program.	
The system shall display the location where the internship takes place for each individual program.	
The system shall display the skills required for each individual program.	
The system shall display if the internship program is on-site, remote or hybrid for each individual program.	
The system shall display the application deadline for each individual program.	
The system shall display details about the salary (if it is a paid internship and option to continue working there) for each individual program.	
The system shall display reviews for companies from verified students that have worked there before.	
The system shall display the knowledge you'll acquire by completing the internship.	
The system shall allow the students to use search filters (location, skills, domain, duration, education level, university partnership).	
The system shall allow users to see internships programs based on the skills and interest of each student.	
The system shall provide the students with status updates for their applications (delivered, read, answered).	
The system shall provide students with interview preparation resources.	
The system shall provide the relevant notifications (application status, internship options) via email notifications.	
The system shall display a tracker showing the number of people applying for the same position concurrently.	
The system shall allow the user to set the frequency they wish to receive updates for their applications.	
The system shall provide contact information for each company.	
The system shall display if a company provides accommodations if the internship is in a different city.	
The system shall provide the option to filter out the unrelated internship programs based on the student's specialization.	
The system shall not allow the student to apply for unrelated internship programs.	
The system shall allow the student to customize their profile (setting their university, specialization, personal details, upload CVs, set their interests)	
The system shall allow the students to automatically apply to any eligible internships using their profile information. (the system shall ask for any extra documents if the internship requires them)	
The system shall provide AI matching between students and companies (comparing student's profile with company requirements and providing the best matchups)	
The system shall allow students and companies to contact each other directly via DMs, telephone numbers or email	
The system shall keep and display a history of all applications	
The system shall allow the user to save internships	
The system shall look professional by not having striking colors or flashy ads	
The system shall display the internship categories in a sidebar, as that is most intuitive	
The system shall be available in browser both for PC and mobile	
The system shall allow a user to compare two internships side to side	
----- Introspection ↓	
The system shall allow companies to register.	
The system shall allow a company to post multiple internship programs.	
The system shall allow a company to see the applicants for one of their internship programs.	
The system shall allow a company to read in detail the applications. (see CV, details about the student that applied)	
The system shall allow a company to accept or reject an application and give feedback.	
The system shall allow a company to edit or remove an internship program.	
The system shall allow a company to get in contact with a student.	
The system shall allow a company to get in contact with a university.	
The system shall allow a company to post pictures on their page.	
The system shall allow a company to specify what student's specializations can take part in the internship	
----- Introspection ↓	
The system shall allow a university to register.	
The system shall allow a university to get in contact with a company.	
The system shall allow electronic filling and signing of contract paperwork between all 3 parties.	

- There were only 3 relatively critical bugs discovered. Whereas the number of bugs is not big, they represent significant problems within the app and the app can not be comfortably used without their fixing.

Bugs identified
1. Can filter by negative Salary
2. An internship can be added with no data
3. Can sign-up a company with no data (no description or even no email or password)

- So this app can NOT be deemed as functional, due to the lack of important features implemented and the existence of critical bugs.

Conclusions

This was a very time consuming process. Especially since I started on a different project and had to move to this project half way, due to a lack of well functioning tools for maintainability metrics. But overall I am pleased with the results, it offered me a good insight into what the next steps for the

development of this app should be. For the future I will continue to do usability testing, as I believe it to be one of the more important software quality factors, but I will probably refrain from using measuring tools as the free versions tend to lack in both functionality and documentation.